

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z3,Z4

Generated by MacAnalog-Symbolix

October 17, 2025

Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple **Z3 Z4:** $\frac{Z_3Z_4g_m}{2Z_3g_m+Z_4g_m}$

$$H(z) = \frac{Z_3Z_4g_m}{2Z_3g_m + Z_4g_m}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3R_3s+1}, R_4 + \frac{1}{C_4s}, \infty, \infty\right)$

$$H(s) = \frac{C_4R_3R_4s + R_3}{C_3C_4R_3R_4s^2 + s(C_3R_3 + 2C_4R_3 + C_4R_4) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}}{C_3R_3+2C_4R_3+C_4R_4}$
wo: $\frac{1}{\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}}$
bandwidth: $\frac{C_3R_3+2C_4R_3+C_4R_4}{C_3C_4R_3R_4}$
K-LP: R_3
K-HP: 0
K-BP: $\frac{C_4R_3R_4}{C_3R_3+2C_4R_3+C_4R_4}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3s}, \frac{R_4}{C_4R_4s+1}, \infty, \infty\right)$

$$H(s) = \frac{C_3R_3R_4s + R_4}{2C_3C_4R_3R_4s^2 + s(2C_3R_3 + C_3R_4 + 2C_4R_4) + 2}$$

Parameters:

Q: $\frac{2\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}}{2C_3R_3+C_3R_4+2C_4R_4}$
wo: $\frac{1}{\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}}$
bandwidth: $\frac{2C_3R_3+C_3R_4+2C_4R_4}{2C_3C_4R_3R_4}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_3R_3R_4}{2C_3R_3+C_3R_4+2C_4R_4}$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (\infty, \infty, R_3, R_4, \infty, \infty)$

$$H(s) = \frac{R_3 R_4}{2R_3 + R_4}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(\infty, \infty, R_3, \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{R_3}{2C_4 R_3 s + 1}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(\infty, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty\right)$

$$H(s) = \frac{R_3 R_4}{2C_4 R_3 R_4 s + 2R_3 + R_4}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(\infty, \infty, R_3, R_4 + \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{C_4 R_3 R_4 s + R_3}{s(2C_4 R_3 + C_4 R_4) + 1}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, R_4, \infty, \infty\right)$

$$H(s) = \frac{R_4}{C_3 R_4 s + 2}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{1}{s(C_3 + 2C_4)}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty\right)$

$$H(s) = \frac{R_4}{s(C_3 R_4 + 2C_4 R_4) + 2}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{C_4 R_4 s + 1}{C_3 C_4 R_4 s^2 + s(C_3 + 2C_4)}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, \infty, \infty\right)$

$$H(s) = \frac{R_3 R_4}{C_3 R_3 R_4 s + 2R_3 + R_4}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{R_3}{s(C_3 R_3 + 2C_4 R_3) + 1}$$

9.11 X-INVALID-ORDER-11 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{R_3 R_4}{2R_3 + R_4 + s(C_3 R_3 R_4 + 2C_4 R_3 R_4)}$$

9.12 X-INVALID-ORDER-12 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_3 R_3 R_4 s + R_4}{s(2C_3 R_3 + C_3 R_4) + 2}$$

9.13 X-INVALID-ORDER-13 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 R_3 s + 1}{2C_3 C_4 R_3 s^2 + s(C_3 + 2C_4)}$$

9.14 X-INVALID-ORDER-14 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 R_3 R_4 s^2 + s(C_3 R_3 + C_4 R_4) + 1}{s^2(2C_3 C_4 R_3 + C_3 C_4 R_4) + s(C_3 + 2C_4)}$$

10 X-INVALID-WZ

11 X-PolynomialError